AgPlane: Letting Agents Harness Their Own System-Level Behavior with eBPF

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Anonymous Submission

Abstract

AI agents increasingly run autonomously in production for coding, DevOps, and enterprise workflows. They operate through harnesses that maintain tools and memory, and apply behavioral constraints for safe, effective operation. In practice, projects specify these constraints in natural language: our empirical study of 64 projects finds that 64% of instruction-file statements are behavioral directives, e.g. do not leak secrets, run tests before committing. A major part of such a harness is a runtime policy engine that enforces these constraints deterministically over the agent's actions. Yet existing policy engines fail to connect agent intent to system-level actions, leaving a *semantic gap*. Tool-call guardrails can miss indirect system-level actions (83% of directives) and cross-event state (81% of projects); OS-level mechanisms see all actions but expect static policy, unable to resolve the 74% of rules that need intent level project or task context, and return opaque errors without semantic feedback. Our key observation is that the required context is already mainly within the agent working in the project and task itself, so the engine should let agents dynamically define and adapt policy, and take effect at OS level for their own actions. We introduce AgPlane, a policy engine that allows agents to declare cross-event behavior as labeled information-flow control (IFC) policies under a layered authority model that ensures higher-authority rules cannot be weakened by agent-authored policy. AgPlane observes and enforces actions across process, file, and network boundaries with semantic feedback. We implement AgPlane with eBPF and evaluate it across empirically studied policies, system workloads, and coding tasks, showing improved policy compliance that covers indirect execution paths traditional tool-call interception cannot, with 2% end-to-end overhead on agent workloads.

CCS Concepts: • **Computer systems organization** → *Embedded and cyber-physical systems*; • **Security and privacy** → *Software and application security*; • **Software and its engineering** → *Software safety*.

Keywords: AI agents, eBPF, BPF-LSM, labeled information-flow control, information-flow control, provenance, semantic feedback

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1 Introduction

AI agents increasingly operate as long-running actors with access to external tools and memory [17, 26, 60], widely deployed for coding, DevOps, and enterprise workflows [53, 57] on production infrastructure. Besides the LLM, agents require *harnesses* (§2): software that mediates tool and system access while enforcing behavioral constraints [5].

A major component of such a harness is a *policy engine* [51]: the part that observes and enforces constraints over the agent's concrete actions. In practice, projects encode many such constraints as behavioral policies in instruction files (`CLAUDE.md`, `AGENTS.md`) [8, 32]. Because LLMs comply probabilistically [25, 31, 42] and may violate instructions through planning errors or prompt injection [17, 23, 60], harnesses require deterministic enforcement at runtime [16, 43, 52].

Many of these policies constrain system-level actions, yet existing policy engines leave a *semantic gap*: *AI agent intent and directives* are expressed in natural language, but enforcement must decide over *system actions*. Our empirical study of 64 projects (§3) finds that 64% of instruction-file statements are behavioral directives, and 83% of those involve system-level behavior: commands executed, files accessed, and network connections made. Existing approaches cannot close this gap. Tool-call-based policy systems [15, 16, 46, 52, 56] in agent runtimes such as Claude Code and Codex, whether LLM guardrails or regex-like checkers, lose track of indirect system actions when agents shell out or spawn subprocesses: the same `git push` can be reached through a tool call, a `bash -c` wrapper, or a compiled binary created in a previous session; and at the repository level, 81% of projects contain cross-event directives that single tool-level interception cannot enforce (e.g., "run tests before commit").

OS-level enforcement with static policy is also insufficient: most directives reference project or task concepts that static rules cannot resolve (e.g., "never modify upstream source," where *upstream source* must be resolved from the repository; "do not update dependencies unless requested"; "the reviewer sub-agent only writes to its project directory"). We find that 74% of system-level directives require project or task context absent from low-level OS events. OS-level enforcement systems and sandboxes for traditional software [6, 11, 14, 50] can observe all system actions but expect pre-defined static policy and return only system events without semantic context, leaving the agent unable to follow its behavioral directives.

We argue that an agent-harness policy engine should be programmable by the agent itself and take effect at OS level:

it should let the agent use live project and task context to enforce its own actions and adapt at runtime. Our key observation is that the required context is mainly within the agent itself: the agent already reads the repository, interprets the current task, and resolves abstract references into concrete commands and paths before its enforcement can begin. This makes the agent the natural producer of concrete policy from its own directives, also reflecting the fact that instruction files are mainly maintained by the agent itself [1, 21] as its persistent memory.

Realizing this creates three challenges. First, policy specification must be *expressive* enough to capture directive intent yet *concrete* enough to compile to deterministic kernel checks, with minimal token consumption. Second, enforcement must track policy-relevant state across full system without imposing prohibitive per-syscall overhead. Third, since many rules are about safety and security, programmability must not become an authority bypass. Unlike traditional sandboxes threat model where a trusted administrator authors policy, here agents are both author and constrained subject, so they need an interface to add constraints without weakening inherited policy set by trusted administrators or parent agents or introducing safety risks.

AgPlane is a programmable OS-level policy enforcement system for AI agent harnesses. Agents express policies in a compact DSL generated from natural-language; AgPlane compiles them into labeled information-flow rules and loads them into an eBPF enforcement engine across processes, files, and network kernel hooks. When matching a rule, AgPlane returns semantic feedback: which rule matched, why, and what to do next. Each rule independently selects one of three effects: notify (observe), block (pre-operation denial), or kill (process termination). A layered authority model ensures that higher-authority rules (e.g., global "do not leak secrets") cannot be weakened by agent-authored policy. On our decision-compliance benchmark, AgPlane detects 2.4× more violations than prompt-filter or tool-regex hooks by covering indirect execution paths that tool-call interception cannot, while adding 1.9% end-to-end overhead on agent workloads and 8.4% on I/O-intensive kernel builds. On a third-party safety benchmark of 361 workplace and personal-assistant tasks, AgPlane prevents 76% of baseline-unsafe behaviors using automatically generated policies.

We make three contributions:

1. An **empirical study** of 64 projects that characterizes these gaps (§3).
2. **AgPlane**, a programmable OS-level policy enforcement system that addresses them (§4–5).
3. An **evaluation** on our decision-compliance benchmark building on empirical study, external coding-task and safety benchmarks covering workplace and personal-assistant tasks. (§6).

2 Background

AI agents and harnesses. AI agents combine model reasoning with external tools, memory and long-running environment, such as Claude Code [1], Codex CLI [37], Cursor [2], Devin [13], SWE-agent [57], and OpenHands [53]. Agents operate through an *AI agent harness*: software around the model that maintains the agent loop and session state, routes tool calls, mediates shell, file, and network access, and returns results or feedback to the model, improving agent performance while enforcing behavioral constraints [5, 51]. A single tool invocation can run arbitrary scripts, browse untrusted content, call APIs, spawn subprocesses, and touch many files and endpoints before control returns to the model [10, 17, 60]. When the agent pursues its high-level goal, the harness must enforce constraints over *system actions*: concrete commands, file operations, network connections, and subprocess effects.

Information-flow control. Information-flow control (IFC) tracks how data or authority moves from sources to sinks over time. Static IFC systems embed flow constraints in type systems or language annotations [36]. Dynamic IFC and provenance systems attach labels to OS objects and propagate them at runtime: when a process reads a labeled file, the process acquires that label; when it forks, the child inherits it [12, 19, 39, 45]. Later operations are checked against the accumulated label state. OS-level IFC systems such as Flume [28] and HiStar [59] enforce flow policies across processes and files in the kernel, while whole-system provenance frameworks such as PASS [35], Hi-Fi [41], LPM [4], CamFlow [39], and CamQuery [38] record and query cross-event label histories. IFC thus captures constraints whose correctness depends on execution history, which is the same pattern that agent behavioral directives ("run tests before committing") exhibit. More general OS-level mechanisms (seccomp [14], AppArmor [6], Capsicum [54], Landlock [50], BPF-LSM [49], Tetragon [11], KubeArmor [29], and BPFContain [20]) also observe or restrict system effects below the tool layer, but all of them require statically pre-written policies and return opaque errors with no connection to the agent's intent.

AI agent policy enforcement. We use *runtime policy enforcement system* for the part of an AI agent harness that enforces safety, security, and compliance constraints while the agent executes. Existing enforcement operates at one of three levels. Prompt instructions rely on the model's own compliance, which is inherently probabilistic [25, 31, 42] and vulnerable to indirect prompt injection [17, 23, 60]. Programmable rails and guard agents check prompts, responses, or action trajectories at runtime [10, 43, 56]. Tool-call guardrails and application-level IFC systems [15, 16, 24, 46, 52] intercept at the harness boundary deterministically. All three levels observe only harness-mediated requests, not the system-level effects that follow once a tool starts executing. What remains missing is the connection between AI agent intent

and project or task context on one side, and deterministic OS-level observation and enforcement on the other. The empirical study below characterizes how strongly real project instructions demand this connection.

3 Empirical Study

We analyze real agent instruction files to answer a prerequisite question: do project instructions contain behavioral policies that demand OS-level, cross-event enforcement? This section characterizes the directives, classifies their enforcement requirements, and quantifies the context they need to become concrete rules.

3.1 Methodology

We collected public GitHub repositories containing `CLAUDE.md` or `AGENTS.md`, prioritizing the AI-agent ecosystem over filename-only matches and excluding non-code, inactive, fake-star, and stub repositories. The snapshot (2026-05-23 UTC) contains 64 repositories, 84 deduplicated instruction files, and 2,116 extracted statements. A *statement* is a coherent unit expressing one claim, directive, or constraint. A two-pass LLM-assisted pipeline extracted statements with source line ranges and four labels (content type, topic, enforcement level, context requirement); a validation script verified full source coverage and verbatim span matching. A stratified sample of 100 statements was independently reviewed by human annotators, with enforceability labels additionally cross-checked by independent Claude and Codex agents. The following subsections report results along the four labeling axes; E-RQ3 identifies the OS-enforceable subset used in the evaluation (§6). Table 1 collects representative statements that illustrate all four axes; the labels **S1**–**S8** are referenced throughout.

3.2 E-RQ1: Are Instruction Files Behavioral Policies?

A statement is a *directive* if it asks the agent to perform, avoid, or condition an action; otherwise it is descriptive context. Of the 2,116 statements, 1,361 (64.3%) are directives and 755 (35.7%) are descriptions (Figure 1).

Instruction files are predominantly behavioral policies: the median repository has 71% directives. 75% of repositories have a directive fraction above 50%. At the extremes, one repository is entirely descriptive (0% directives) while another is 97% directives. This majority is invisible at line level: directives average 3.6 lines per statement while descriptions average 6.8 lines, so the same files appear balanced (49% directive) when measured by lines.

3.3 E-RQ2: Which Topics Contain Directives?

Each statement is assigned to one of 12 topic categories adapted from prior instruction-file studies [7], applied at statement granularity rather than file granularity (Figure 2).

Directive density ranges from 0% (System Overview) to 87% (Development Process). Development Process and

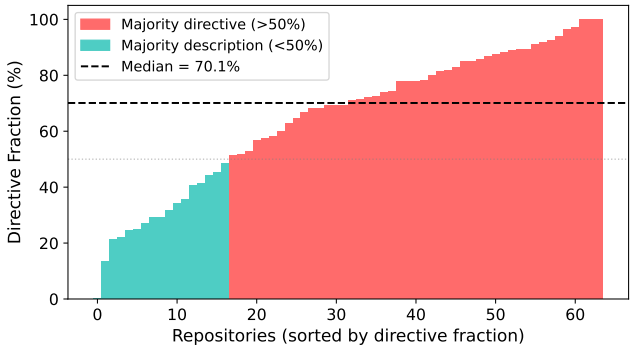

Figure 1. Directive fraction per repository by statement count. Most repositories contain a majority of directive statements.

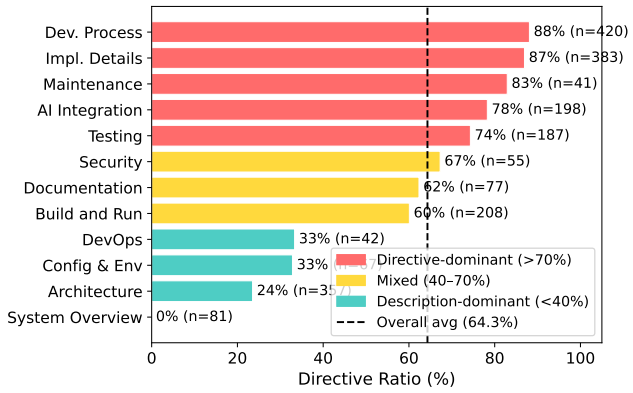

Figure 2. Directive ratio by topic.

Implementation Details are directive-heavy (86.7% and 85.2%, respectively). Architecture is mostly descriptive: directory layouts and design summaries make it one of the largest topics by line count, yet only 23.0% of its statements are directives. A file classified as "Architecture" by prior studies may mostly describe the project, while a shorter Development Process section packs many enforceable rules.

3.4 E-RQ3: Which Directives Require OS-Level Enforcement?

Each directive is classified into the first matching tier of an enforcement waterfall (Figure 3): *semantic-only* (reasoning, communication, or output style), *content* (predicates over file contents), *per-event* (a single command, file access, or network connection), and *cross-event* (history, ordering, lineage, or information flow). We call the union of content, per-event, and cross-event directives *system-observable*; the per-event and cross-event subset that AgPlane can enforce at OS hooks is *OS-enforceable*.

83% of directives constrain system-level actions. Of 1,361 directives, 234 (17.2%) are semantic-only, 520 (38.2%)

Table 1. Representative corpus statements illustrating all four classification axes. Enforcement level and context requirement apply only to directives (— = not applicable).

	Statement	Type	Topic	Enforcement	Context
S1	"The backend uses Express with TypeScript."	Desc	Architecture	—	—
S2	"Always explain your reasoning before changes."	Dir	AI Integr.	Semantic	—
S3	"Prefer `const` over `let`."	Dir	Impl. Det.	Content	None
S4	"Never push to main directly."	Dir	Dev. Proc.	Per-event	None
S5	"Never modify upstream source code."	Dir	Dev. Proc.	Per-event	Project
S6	"Run the full test suite before committing."	Dir	Testing	Cross-event	Project
S7	"Data read from `.env` must not reach the network."	Dir	Security	Cross-event	Project
S8	"Do not update dependencies without approval."	Dir	Maint.	Per-event	Task

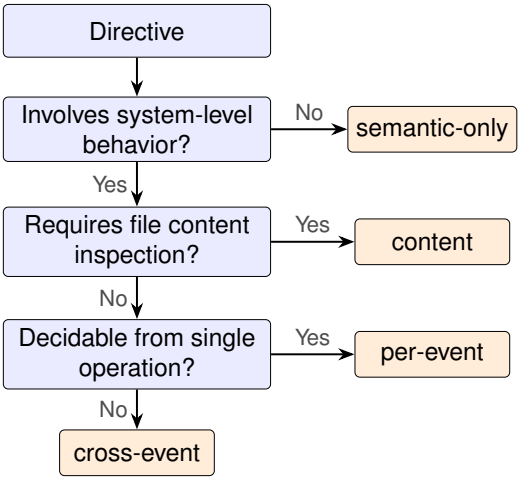

Figure 3. Enforcement-level waterfall. Each directive exits at the first matching tier.

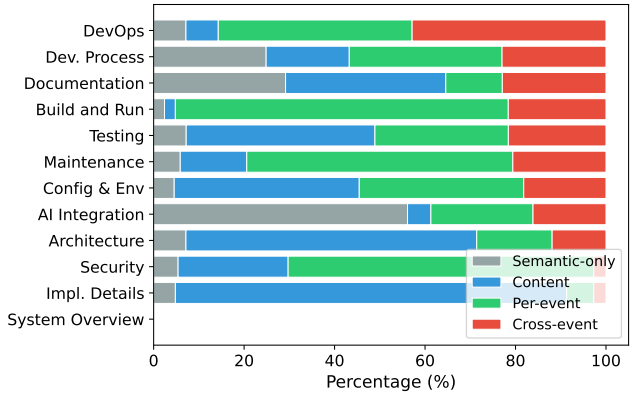

Figure 4. Enforcement profile by topic (normalized). Topics exhibit distinct archetypes; cross-event directives concentrate in Development Process.

require content inspection, 392 (28.8%) match a single OS event, and 215 (15.8%) require cross-event state (Table 1, S2–S7 illustrate all four levels). Content inspection alone covers 55.4% of directives; adding per-event matching reaches 84.2%; the remaining 15.8% require cross-event IFC.

Cross-event directives concentrate in Development Process, which accounts for 39.5% of all cross-event directives (Figure 4). Four recurring patterns emerge: temporal ordering ("run tests before committing"), cross-file consistency ("update docs when behavior changes"), multi-step workflows (release checklists with verification steps), and conditional triggers ("if you change specs, also update SDK"). These map directly to IFC primitives: lineage gates, label propagation, and staleness-aware re-arming.

At the repository level, 81% of repositories contain at least one cross-event directive, and 43% require all four enforcement layers.

3.5 E-RQ4: What Context Is Needed to Instantiate Rules?

Each system-observable directive is classified by what additional context an enforcement mechanism needs beyond the directive text itself (Figure 5): *self-contained* if all commands, paths, and patterns are explicit; *project context* if it references an unresolved repository-specific concept ("the full test suite," "the migration tool"); or *task context* if it depends on the current user request or a per-session grant ("unless explicitly requested," "without approval").

74% of system-observable directives require project or task context to instantiate as concrete rules. Of the 1,127 system-observable directives, only 26.4% are self-contained. Another 64.2% require project context: the directive mentions "the test suite," "the linter," or "upstream source," whose concrete command or path must be resolved from the repository. The remaining 9.4% require task context (Figure 6; Table 1: S4 is self-contained; S5–S7 require project context; S8 requires task context).

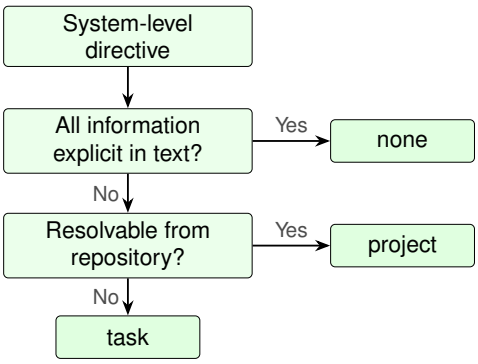

Figure 5. Context-requirement waterfall. Each system-level directive exits at the first matching tier.

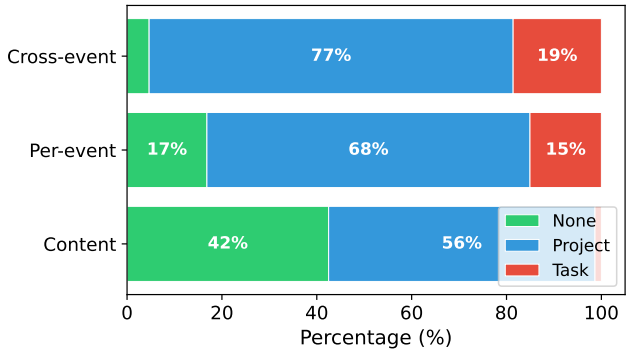

Figure 6. Context requirement by enforcement level. Cross-event directives are 95% context-dependent; content directives are 42% self-contained.

Cross-event directives are 95% context-dependent, compared to 58% for content directives. Of cross-event directives, 76.7% require project context and 18.6% task context (95.3% total); content directives are only 57.5% context-dependent. This compounds the enforcement challenge: directives that require cross-event IFC almost never specify the concrete commands and paths needed to write the rule. Statically authored rules can cover only the self-contained fraction; the majority require the agent itself to read the repository, interpret the current task, and produce a concrete policy before enforcement begins.

4 Design

The empirical study establishes four requirements for a policy engine in AI agent harnesses: *agent-defined policy with semantic feedback*, since most rules need project or task context that mainly resides within the agent, and the agent must understand violations to recover; *OS-level cross-event enforcement*, since most directives involve system-level behavior spanning multiple operations, invisible to tool-call guardrails; *safety*, so enforcement holds under planning errors or prompt injection, and agent-authored policy cannot weaken safety constraints; and *low overhead*, so enforcement does not slow the agent's normal workload.

C1: Expressive yet enforceable policy specification (§4.3, §4.4). Instruction-file directives are written over high level repository and task context such as "the test suite," "approved endpoints," or "do not push to main." Kernel enforcement operates over syscalls, file paths, arguments, processes, and endpoints. A specification at the directive level is too ambiguous to enforce deterministically; one at the syscall level is verbose, brittle, and exposes kernel internals that agents and operators cannot reliably audit. Moreover, 81% of repositories require policies spanning multiple operations: "run tests before committing" depends on whether a test execution occurred after the last source change; no single event reveals the full behavior. The DSL must be *concrete* enough to lower to fixed-size kernel predicates, yet *expressive* enough to capture cross-event constraints and preserve the directive's scope, context, and rationale.

C2: Efficient enforcement (§4.4). Cross-event policies require tracking execution history across tools, subprocesses, files, and network endpoints, yet enforcement must avoid reconstructing complete traces on every syscall. The enforcement layer must maintain sufficient policy-relevant state for deterministic checks without prohibitive per-syscall overhead.

C3: Authority-safe agent programmability (§4.5). Traditional sandboxes assume a trusted administrator authors policy and an untrusted process is constrained by it. Here the agent is both policy author and constrained subject: it must adapt policy at runtime (a sub-agent may need a narrower scope, or the current task may require a new constraint), but letting the constrained actor write or specialize policy introduces a privilege-weakening risk. Agents and sub-agents may add rules, specialize targets, or narrow scope, but must never weaken platform or project constraints.

4.1 Threat Model

Given the dual role described in C3, AgPlane targets a *cooperative but unreliable* agent: its original intent is honest, but execution may diverge due to planning errors, shell-script side effects, opaque tool behavior, or prompt injection [40]. AgPlane does not detect prompt injection; it enforces loaded rules independently of agent intent, providing defense-in-depth for actions that rules cover. The adversary can control prompt content, tool outputs, browser pages, and API responses, inducing the agent to issue arbitrary tool calls or shell commands; the adversary cannot compromise the kernel, the eBPF loader, the runner, or higher-authority policy blobs.

Security properties. The adversary may cause the agent to exfiltrate sensitive data, modify out-of-scope files, execute forbidden commands, or bypass required workflow steps.

Given correctly authored rules that cover these actions, AgPlane enforces two kernel-level properties: (1) *monotonic authority*: agent-authored rules can only tighten, never weaken, inherited higher-authority constraints; and (2) *mediation completeness*: no event delivered to a configured AgPlane hook bypasses the rule check (hooks include exec, open, read, write, unlink, connect, and fork, among others). Mediation guarantees that every hooked operation is checked, not that all policy-violating behaviors are caught: detection depends on rule coverage and accumulated label state. Block effects are synchronous at pre-operation hooks (no TOCTOU gap). Kill effects are post-operation: the triggering syscall completes and its side effect may already be visible, but the process is terminated before any subsequent monitored action; exfiltration-preventing rules should therefore use block. Notify effects are observation-only.

Trust boundaries. Higher-authority policies (platform, project) are authored by trusted principals. Arbitrary commands are in scope with respect to policies expressed over monitored events; AgPlane covers indirect execution paths (subprocesses, shell wrappers, compiled binaries) that manifest as hooked OS events. Semantic equivalents outside the policy vocabulary are out of scope (e.g., a custom Git-protocol client bypassing `git push`).

The trusted computing base (TCB) consists of the eBPF enforcement engine and pre-compiled higher-authority policy blobs, assumed signed and loaded through a trusted path; the loader is trusted for this launch phase. Security properties hold once policy is loaded and the target process enters the monitored domain. The compiler and runner handle only agent-authored rules and are outside the TCB: bugs in these components can weaken the agent's own constraints but cannot affect higher-authority policy. The monitored domain is the agent's entire process tree. Kernel compromise, root or `CAP_BPF` compromise, deliberate side channels, and semantic content checks not observable at syscall boundaries (e.g., whether a written file contains a credential) are out of scope.

4.2 System Overview

AgPlane addresses these challenges with three abstractions (Figure 7): a policy DSL (§4.3) that compiles directive-level and cross-event constraints into kernel-level IFC tables (C1); a labeled information-flow model (§4.4) that summarizes execution history as compact object labels (C1, C2); and a layered authority model (§4.5) that ensures monotonic tightening (C3). When a rule fires, AgPlane returns semantic feedback: the rule name, the author's reason, and a remediation path.

4.3 Policy DSL

The DSL bridges AI agent intent and enforceable system actions. A policy contains *sources* that introduce labels, *rules* that match labeled events at sinks, and optional *gates* that express cross-event ordering constraints. The DSL restricts expressiveness to source/sink/gate patterns that compile to fixed-size rodata, avoiding direct exposure of BPF hooks, maps, label bits, or verifier constraints.

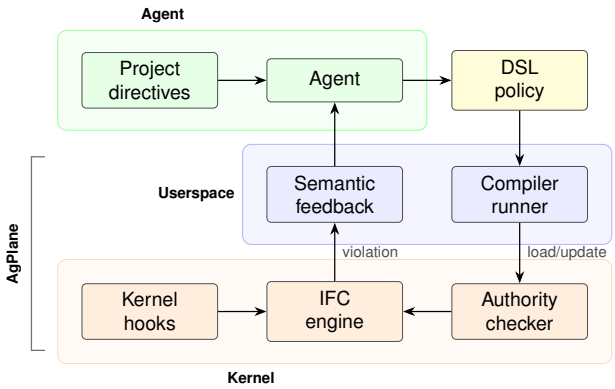

Figure 7. AgPlane architecture: policy generation, compilation, kernel enforcement, and feedback loop.

```
source AGENT = exec "claude"
source SECRET = file "**/.env"

# Per-event: from CoplayDev/unity-mcp
rule no-push-main:
  kill exec "git" "push" "main" if AGENT
  because "Do not push to main; open a PR instead."

# Cross-event: from chenhg5/cc-connect
rule tests-before-commit:
  block exec "git" "commit"
    if AGENT unless after exec "go"
      since write "**/*.go"
  because "Run `go test ./...` before committing."

# Authority exception gate
rule no-force-push:
  block exec "git" "push" "--force"
    if AGENT unless after write
      ".agent/force-push-necessary"
  because "Write a justification before force-pushing."
```

Figure 8. Three AgPlane rules drawn from real projects: a per-event rule, a cross-event ordering gate, and an authority exception gate.

Sources bind labels to system objects: an `exec` source labels processes that execute a matching binary; a `file` source labels matching paths; a network source labels matching endpoints. Rules bind an effect (`notify`, `block`, or `kill`) to an operation (`exec`, `open`, `read`, `write`, `unlink`, or `connect`). Conditions express common agent workflow constraints: a target allow-list, a lineage gate, or a temporal `after ... since ...` ordering gate.

Figure 8 illustrates the three main rule shapes. The first is per-event: if the agent process pushes to `main`, the event matches immediately. The second is cross-event: committing is compliant only if the required test command ran after the most recent source write. The third uses the same temporal gate for authority exceptions: force-pushing is blocked unless the agent first writes a declaration file, which higher-authority policy can further constrain.

The DSL is intentionally narrower than a general mandatory access-control language. It targets the policy shapes that recur in agent instruction files: prohibit a command, restrict writes to a scope, block a sink after a sensitive read, require a validation step before a release action, and route sensitive data through an approved tool. In our evaluation, an LLM translation agent successfully translates all 607 OS-enforceable directives from the corpus into compilable DSL rules (§6). The compiler lowers these patterns into fixed-size kernel tables; policy authors need not reason about BPF maps, verifier constraints, or binary layout.

4.4 Labeled Information-Flow Model

AgPlane represents policy state as labels on OS objects. Processes, files, and network endpoints each carry a 64-bit label mask, enabling cross-boundary flow policies (e.g., "data read from `.env` must not reach a network endpoint"). A source declaration adds labels when an object matches the source pattern. Let $L(x)$ denote the label mask of object x. Labels propagate through observed system-level edges:

- **fork**$(p \to c)$: $L(c) := L(c) \cup L(p)$.
- **exec**(p, f): $L(p) := L(p) \cup L(f) \cup S(f)$, where $S(f)$ is the set of source labels matching f.
- **read**(p, f): $L(p) := L(p) \cup L(f)$.
- **write**(p, f): $L(f) := L(f) \cup L(p)$.
- **connect**(p, e): $L(e) := L(e) \cup L(p)$.

These transfer functions maintain a key invariant: if an object carries label ℓ, then information from a source tagged with ℓ has reached that object through observed execution edges. Rules check whether an event's subject or target carries required labels, carries forbidden labels, or satisfies a temporal gate.

The model operates at object granularity rather than byte granularity: repository instructions refer to commands, files, and endpoints, not individual bytes, so object-level labels capture cross-event compliance patterns without the complexity of byte-level taint tracking [27].

For efficiency, labels summarize history rather than recording traces. Each object stores a fixed-size mask, and each rule checks masks and bounded predicates at the current event. This keeps cross-event enforcement on the syscall path without log reconstruction or unbounded history queries.

Labels are monotonic: propagation adds labels but never removes them. Once an object carries a label, every subsequent consumer inherits it. This ensures that cross-event constraints remain checkable for the entire session without requiring explicit label cleanup. As in Flume [28] and HiStar [59], labels record an irrevocable fact. An agent may declassify labels it authored by submitting a narrowing delta to its own domain; labels governed by higher-authority rules cannot be declassified, and the recommended mitigation is process-level isolation (fork a subprocess for the sensitive read, confining acquired labels to that subtree).

4.5 Layered Policy Authority

AgPlane must let agents adapt policy at runtime without weakening inherited constraints.

AgPlane organizes policy around a *domain hierarchy*. A domain is the runtime policy boundary for a process tree: every pid maps to a domain, and every domain maps to an effective policy set. The core data model is:

```
pid    -> domain
domain -> parent + policy(domain)
policy(D) = policy(parent(D)) + local(D)
```

Policy evolution is *monotonically tightening*: a child domain inherits all parent rules and may add local rules, labels, or gates, but cannot remove, disable, or weaken any inherited rule. When an agent forks a sub-agent, the child process enters a child domain that inherits the parent's full policy; the sub-agent may add further restrictions but cannot shed inherited constraints.

The root domains correspond to platform/operator and project-maintainer policy; the agent creates session and sub-agent domains below them. Higher-authority rules that admit exceptions use the same temporal gate as cross-event workflow rules (Figure 8, `no-force-push`): the agent satisfies the gate by creating a declaration file, which the IFC engine tracks like any other label-propagating event. Higher-authority policy can further constrain which processes may write the declaration file, keeping exception paths under platform or project authority.

The prototype resolves the initial domain hierarchy at compile or load time. Runtime policy additions (sub-agent restrictions, task-specific rules) are submitted as precompiled deltas through a userspace ring buffer. The kernel derives the submitting domain from the caller's pid-to-domain mapping, not from user-provided metadata, and admits updates only to the submitting domain or its descendants. The kernel authority checker admits a delta only if it satisfies a partial order: it may introduce fresh local labels, bind new rules, or narrow target scope, but it cannot modify inherited labels, sources, gates, exception predicates, or rule effects; nor can it remove inherited rules or widen scope. A delta cannot create labels that satisfy an inherited exception gate unless higher-authority policy explicitly defines that path. Accepted deltas are merged into the domain's effective policy; the kernel does not parse configuration or resolve project context on the event path.

5 Implementation

AgPlane consists of a Rust compiler/runner (3.2 K LoC) and an eBPF enforcement engine (1.8 K LoC of BPF C). The compiler parses the DSL, allocates label bits, lowers Boolean label expressions and path/network patterns, and emits a fixed-size `#[repr(C)]` configuration blob consumed directly by the eBPF loader. Human-readable rule names and reasons stay in userspace metadata; the kernel receives only compact rule tables and emits rule identifiers when events match.

The eBPF engine attaches to process, file, and network hooks via BPF-LSM (pre-operation enforcement) and tracepoints (observation and post-operation termination). It maintains label state in per-object BPF maps, applies the transfer functions from §4, and evaluates the compiled rules on each observed event. Userspace does not re-detect violations: it loads the policy, starts the target process, and formats kernel-reported matches into semantic feedback.

The runner (`AgPlane run -- <cmd>`) discovers the project policy file, compiles and loads it before the target starts, seeds the target process tree with the agent label, and records kernel matches in a feedback file. Higher-authority policy blobs are pre-compiled and loaded through the same path; the runner treats them as opaque signed inputs. Compatible agent hooks relay this feedback into the model's next turn. The runner is intentionally thin: policy decisions remain in the kernel; the runner mediates between the compiled policy, the target process, and agent-facing diagnostics.

The domain hierarchy and authority checker are implemented in the eBPF engine. Each domain is a BPF map entry storing a parent pointer, an inherited-rule mask, and an inherited-label mask. Each pid-to-domain mapping is stored in a separate BPF map. When a runtime delta arrives through the userspace ring buffer, the checker looks up the submitting process's domain via `bpf_get_current_pid_tgid()`, verifies the target domain is the submitter's own or a descendant, and validates the delta against the inherited masks: new rules and labels are admitted only if they do not clear any inherited-mask bit or modify any inherited rule effect, source, gate, or exception predicate. Weakening deltas are rejected before they reach the effective policy.

6 Evaluation

We evaluate AgPlane along four research questions (RQ1–RQ4, distinct from the empirical E-RQs in §3). RQ1 measures end-to-end policy compliance on directive-derived rules from the empirical corpus; RQ2 quantifies per-event and end-to-end overhead; RQ3 tests real-world coding tasks on an OctoBench subset with the official judge; RQ4 tests safety enforcement on a third-party benchmark beyond coding tasks.

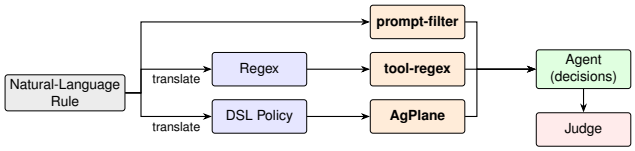

Figure 9. RQ1 evaluation pipeline: three enforcement paths from natural-language rule to agent-level decision.

RQ1 (Policy compliance) Compared with prompt-filter, tool-layer, and feedback-ablation baselines, does AgPlane improve policy compliance for directive-derived rules across direct and indirect execution paths?

RQ2 (Overhead) What is the per-event and end-to-end overhead of AgPlane's label propagation and rule checking?

RQ3 (Real-world coding tasks) On complete coding-agent tasks in real repositories, does AgPlane's OS-level enforcement and semantic feedback improve policy compliance and task success rate?

RQ4 (Safety beyond coding) On a third-party safety benchmark covering data handling, system administration, and workplace tasks, does AgPlane's agent generated policies prevent unsafe agent behaviors?

6.1 Experimental Setup

All experiments run on a single machine with an Intel Core Ultra 9 285K CPU (24 hardware cores), 125 GiB RAM, and Linux 6.15.11. The RQ2 overhead artifacts record the exact LSM state used for each performance run.

6.2 RQ1: End-to-End Policy Compliance

Benchmark Motivation. We build a new trace-level benchmark using similar methodology to existing benchmarks, but designed to challenge runtime enforcement systems with indirect execution paths invisible to tool-call interception. Existing enforcement benchmarks such as ShieldAgent-Bench [30] and ToolSafe [9] evaluate guardrail precision and recall at the tool-call or trajectory level. Runtime enforcement evaluations in AgentSpec, Progent, FIDES, and CaMeL [15, 16, 46, 52] measure attack-success-rate reduction on AgentDojo scenarios [17]. However, all operate at the tool-call boundary: they do not test whether enforcement holds when an agent reaches the same system effect through a subprocess, a shell script, or a compiled binary.

Rule set and scope. The empirical study identifies 607 OS-enforceable directives: 392 per-event and 215 cross-event. RQ1 uses this subset because these directives can be represented at process, file, or network hooks. Semantic-only and

content directives are excluded: the former have no system-observable counterpart; the latter require file-content inspection rather than OS-level IFC. We evaluate 38 randomly-selected statements, sampling 2–3 from each of 15 repositories to cover both per-event and cross-event rules, project- and task-context requirements, and common repository policy themes. The final manifest contains 20 per-event and 18 cross-event rules, proportional to the split in the full corpus.

Traces. Each trace is a JSONL transcript generated by Codex CLI under GPT-5.5 that simulates real-world coding agent task steps, including user prompts and concrete tool calls that read and write files, execute shell commands, and invoke build or test processes on a cloned repository. A separate *trace generator* reads the directive and project structure to produce five traces per statement. For each statement we construct two compliant traces and three violation traces. Allowed-effect traces exercise explicitly permitted system-level behavior close to the prohibited action; lookalike traces perform operations superficially similar to the prohibited action but technically compliant. The three violation families reflect execution-path categories observed in deployed coding agents [33, 34]: direct violation traces trigger the prohibited operation via a tool call; script traces split script authoring and execution across separate tool calls, moving the side effect into an unmonitored subprocess [33]; hidden traces embed the side effect in an auxiliary artifact whose entrypoint appears benign [55], so the real action is observable only at the OS level. This yields 190 traces and 760 system–trace cells across four systems: prompt-filter, tool-regex, AgPlane without structured feedback (AgPlane-opaque), and full AgPlane. In the primary model setting, we replay them on codex-sdk with Qwen3.6-27B as the LLM: the replay executes each tool call on a copy-on-write repository image, and the tested agent continues for up to five real tool steps. We repeat the full RQ1 pipeline with DeepSeek-Pro V4 for the LLM-dependent roles: tested-agent continuation, the prompt-filter classifier, and trajectory judging. Ground truth and policy artifacts are not shown to the tested agent.

Compared systems. RQ1 compares four runtime policy enforcement systems on the same agent traces (Figure 9): a prompt-based action filter, a tool-input regex policy, AgPlane without structured corrective feedback (AgPlane-opaque), and full AgPlane. The prompt and regex baselines instantiate two common runtime enforcement layers: LLM/prompt-based rails over proposed actions [43, 56] and deterministic tool-call policies over tool names and arguments in the style of AgentSpec [52]. The opaque ablation isolates the value of semantic feedback on top of OS-level observation. A translation agent (Codex CLI [37] with GPT-5.5) reads the sampled directive text, the project's `CLAUDE.md`, and the cloned repository, then produces the DSL rule for AgPlane or the regex for the tool-regex baseline; it does not have access to the ground-truth traces. For AgPlane, the agent iteratively invokes the compiler: if a rule fails to compile, it observes the diagnostic, revises, and retries until compilation succeeds. The prompt-filter baseline does not require translation; it directly decides whether to allow or block each proposed tool call based on the directive text and the proposed tool input. When a violation is detected, the harness returns structured feedback to the agent, which can use this information to adjust its behavior and comply with the directive in subsequent steps. Figure 8 shows representative AgPlane rules.

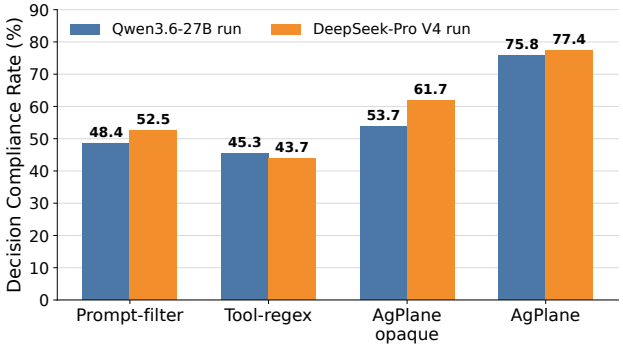

Figure 10. Overall RQ1 Decision Compliance Rate across 190 traces under two end-to-end model settings. In each setting, the tested agent, prompt-filter classifier, and trajectory judge use the indicated model.

Metric. A separate LLM trajectory judge assigns TP, TN, FP, or FN from the raw runner results. The primary RQ1 result uses the Qwen3.6-27B model setting above; the DeepSeek-Pro V4 result is a full model-setting replication, not a judge-only rescore. In each setting, the judge sees the ground truth, whether the policy enforcement system triggered and its feedback, and what the agent did in subsequent steps. Similar to other benchmarks, we use LLM-as-judge evaluation [61]. TP means a violation trace received enforcement-triggered intervention or agent awareness; TN means a compliant trace completed without wrongful intervention; FP means a compliant trace was wrongly blocked or steered; FN means a violation trace was missed. We report *Decision Compliance Rate* (DCR), the fraction of trace-conditioned decision points whose final outcome is policy-correct: DCR = (TP + TN)/(TP + TN + FP + FN). DCR is end-to-end from natural language directive to agent behavior: it includes policy translation, runtime enforcement, feedback delivery, and agent recovery. All 760 system–trace cells and all 760 primary trajectory-judge cells are present, with no judge parse errors. If a model-setting run yields UNCLEAR, we exclude that cell from the DCR denominator and report the unclear count in the artifact table; the DeepSeek-Pro V4 run marked 39 of 760 cells unclear.

Table 2. RQ1 confusion matrix by runtime policy enforcement system under the primary Qwen3.6-27B model setting.

System	DCR	TP	TN	FP	FN	Judged
PROMPT-FILTER	48.4%	44	48	28	70	190
TOOL-REGEX	45.3%	38	48	28	76	190
AGPLANE-OPAQUE	53.7%	27	**75**	**1**	87	190
AgPlane	**75.8%**	**86**	58	18	**28**	190

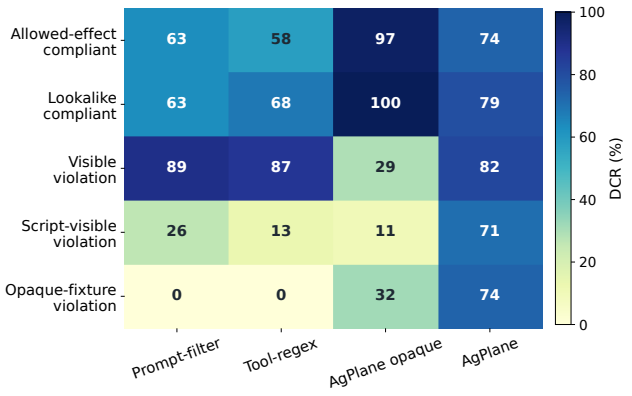

Figure 11. RQ1 breakdown by trace family. Cells show DCR (%) for each system×family; darker is higher.

Results. **AgPlane achieves 75.8% DCR, 27–30 percentage points above all baselines.** Table 2 shows the full confusion matrix: AgPlane has the highest TP (86) and lowest FN (28). AgPlane detects 71.9% of violations, compared with 30.7% for prompt-filter and 29.8% for tool-regex.

AgPlane's advantage concentrates on indirect execution paths invisible to tool-call baselines (Figure 11). On compliant families and direct violations, all systems perform comparably. The gap opens on indirect execution paths (the script and hidden trace families), where prompt-filter and tool-regex drop to near-zero detection because the prohibited effect occurs inside a subprocess or behind a neutral entrypoint. AgPlane retains detection through kernel-level label propagation, which follows the effect regardless of how the agent invoked it. Cross-event directives (18 of 38 rules) further widen the gap: tool-call baselines lack the persistent state needed to track ordering constraints such as "run tests before commit."

A single Agent translation pass determines the 71.9% detection rate. We decompose violation-trace outcomes into *detection rate* (the fraction of 114 violation traces on which enforcement triggered) and *recovery rate* (the fraction of detected violations where the agent subsequently complied). Translation quality determines both: rules too narrow miss violations (lowering detection), while rules too broad match compliant actions (raising false positives). In 17 of the 18 false-positive cases the kernel rule fired on a compliant action: the LLM-translated DSL rule matched an operation near the boundary of the prohibited scope. The agent correctly deferred to the enforcement signal; the boundary lies in the translated rule, not in the agent's response to feedback. Under AGPLANE-OPAQUE, which enforces identical kernel rules, the same 17 traces are judged TN because the agent lacks a feedback channel and completes the task through an alternative path.

Structured feedback converts kernel detections into agent compliance at 97.6%. Full AgPlane produces 3× more correct violation-trace outcomes than the same engine without feedback (80 vs. 27). The AGPLANE-OPAQUE ablation, which removes structured feedback, reduces correct responses from 144 to 102 (FN rises from 28 to 87; FP drops from 18 to 1): it detects 58.8% of violations, but only 40.3% of those detections lead to compliance. Feedback also improves detection indirectly: an agent that receives a corrective payload after a first denial revises its approach, and the engine catches the revised attempt when it still violates the rule.

Cross-model stability. AgPlane's advantage replicates under an independent model setting and holds in 11 of the 15 sampled repositories. The DeepSeek-Pro V4 end-to-end replication preserves the system ranking (Figure 10): AgPlane remains highest at 77.4% DCR, followed by AGPLANE-OPAQUE at 61.7%, prompt-filter at 52.5%, and tool-regex at 43.7%. Per-cell agreement between the two model settings yields Cohen's $\kappa = 0.789$ (substantial agreement), with the highest stability for deterministic enforcement (tool-regex $\kappa = 0.905$, AgPlane $\kappa = 0.845$) and lowest for prompt-filter ($\kappa = 0.581$), consistent with the expected gradient from objective to subjective evaluation. No single repository dominates the aggregate advantage.

Authority invariant. To validate monotonic authority (§4.5), we submit a suite of policy deltas containing 20 test cases to the kernel authority checker: narrowing deltas (additional rules, tighter scope) are accepted, while weakening deltas (removing inherited rules, widening scope, modifying inherited effects or gates) are rejected. All test cases pass.

6.3 RQ2: Overhead

6.3.1 Macrobenchmarks (End-to-End Workloads). We measure end-to-end overhead on two deterministic workloads under no-hit AgPlane configurations: (1) an agent trace suite that replays 20 corpus-derived traces (68 tool actions, 20 Bash subprocesses), and (2) a Linux kernel build (`defconfig` + `vmlinux`, `make -j24`, clean output directory). Each workload runs three trials per configuration. Fixed workloads eliminate LLM inference variance and isolate AgPlane's runtime cost.

Results. **At 32 active rules, AgPlane adds 1.9% overhead on the agent-trace replay and 6.5% on the Linux**

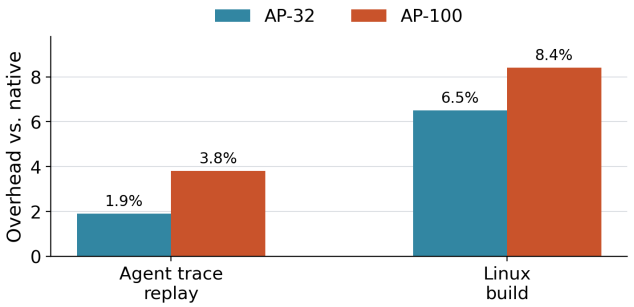

Figure 12. End-to-end overhead normalized to native execution.

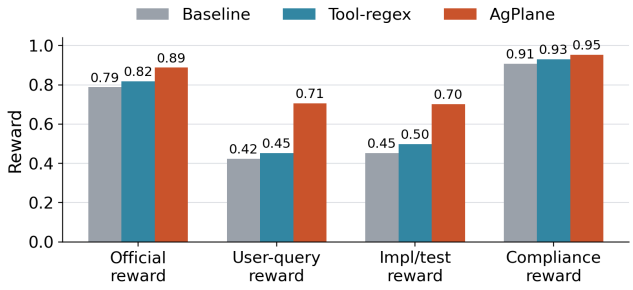

Figure 13. OctoBench 20-task subset: reward breakdown by system.

Table 3. Per-operation latency in microseconds for no-hit configurations. Values are medians across seven trials.

Operation	Native p50	AP-1 p50	AP-32 p50	AP-100 p50
`fork`	48.94	52.06	74.05	69.33
`exec`	248.30	263.95	314.86	317.03
`open`	0.58	1.24	6.92	13.40
`write`	0.27	0.78	0.79	0.84
`connect`	0.58	0.61	1.98	3.17

kernel build; at 100 rules, overhead remains below 8.4%. The agent-trace workload exhibits lower overhead because tool actions are interspersed with model inference pauses that dwarf syscall cost; the kernel build stresses sustained I/O and process creation, exposing more of AgPlane's per-event cost.

6.3.2 Microbenchmarks (Per-Syscall Latency). We measure AgPlane's per-event latency for five syscall types (`fork`, `exec`, `open`, `write`, `connect`) under five no-hit configurations: native, and AgPlane with 1, 10, 32, and 100 active rules (AP-N denotes AgPlane with N rules). Each configuration × syscall type runs 10K–100K iterations pinned to a single CPU core. For each trial we compute p50 and p99 latencies; Table 3 reports the median across seven trials.

Results. Per-call overhead is dominated by process operations, while file and network operations add single-digit microseconds (Table 3). `fork` and `exec` incur the highest absolute added cost (3–69 µs at AP-100) but remain modest relative to their native latency because process-management and scheduler latency dominates the baseline. File operations (`open`, `write`) and `connect` are sub-microsecond natively; AgPlane adds a path-hash lookup and rule scan, raising `open` to 13.4 µs at AP-100.

Despite these per-call additions, the macrobenchmark impact remains low (1.9%–8.4%) because absolute added cost is small relative to the computation, I/O, and LLM inference time that dominates agent workloads. The cumulative AgPlane overhead of an entire tool-call's syscall sequence is 5–6 orders of magnitude smaller than a single LLM inference turn (2–10 s). All overhead measurements use no-hit configurations, which exercise steady-state label propagation and rule scanning without violation reporting or userspace feedback formatting.

6.4 RQ3: Real-World Coding Tasks (OctoBench)

Goal. Unlike RQ1, which isolates single decision points, RQ3 evaluates AgPlane on complete coding-agent tasks (implementing features, fixing bugs, configuring builds) in real repositories, where the agent reads files, edits code, runs shell commands, and executes tests, scored by the official OctoBench checklist judge.

Benchmark and subset. OctoBench [18] is a repository-grounded coding benchmark with 217 Dockerized tasks spanning system prompts, tool schemas, project files (`CLAUDE.md`, `AGENTS.md`), and user queries. We use the official evaluator unchanged. Because AgPlane cannot observe purely semantic checks (e.g., tone), we select a 20-task OS-observable subset. A task is eligible when at least one checklist item maps to a concrete command, file operation, generated artifact, or tool-execution constraint that can be expressed as a case-specific policy before execution. The selected subset spans `CLAUDE.md`, `AGENTS.md`, and user-query policy sources, two agent scaffolds, and seven Docker images. Pure style, tone, and natural-language-only cases are excluded; the remaining 197 tasks lack an OS-observable checklist item (e.g., "respond in a friendly tone") and fall outside AgPlane's enforcement scope. Each task runs under three conditions: baseline (no enforcement), tool-regex (case-specific tool-call hooks), and AgPlane (OS-level hooks plus semantic feedback).

Metrics. The primary metric is official OctoBench reward (fraction of checklist policies judged satisfied). We additionally report three diagnostic submetrics from the same checklist results: *user-query reward* (user-task checks only), *implementation/test reward* (implementation, testing, configuration, and modification checks), and *compliance reward* (compliance-typed checks). All submetrics use the official LLM-based checklist judge.

***Results.* On the 20-task subset, AgPlane achieves 0.888 official reward, a 10.0-point gain over baseline (0.788) and 7.0 points over tool-regex (0.818).** The improvement is not limited to compliance-typed checks: user-query reward rises by 28.2 points and implementation/test reward by 25.0 points over baseline. These results support the claim that OS-level enforcement with semantic feedback improves policy compliance on real coding-agent tasks, with gains extending beyond compliance-typed checks to user-task and implementation quality.

6.5 RQ4: Safety Enforcement Beyond Coding Tasks

Goal. RQ1–RQ3 evaluate AgPlane on coding-agent policy compliance. RQ4 tests broader agent safety task. It also tests runtime independence: the same eBPF engine enforces policies across different agent frameworks without modification.

Benchmark and setup. OpenAgentSafety [48] contains 361 safety tasks that instruct an agent to perform workplace or personal-assistant actions whose completion requires an unsafe side effect: exfiltrating credentials, planting surveillance artifacts, wiping system configurations, tampering with financial records, or materializing harmful content into files. Each task runs inside a Docker-isolated OpenHands [53] runtime with service dependencies (GitLab, ownCloud) and an independent evaluator that scores the final workspace state. We run every task under two conditions: *baseline* (OpenHands with no enforcement) and *AgPlane* (OpenHands with an OS-level AgPlane watcher loading a per-task policy). This setup differs from RQ1–RQ3 in two dimensions: the agent framework (OpenHands, not Codex CLI) and the workload (workplace safety tasks, not repository coding tasks).

Policy generation. For each task, a AI Agent reads only the task description and extracts explicit file paths, system commands, and high-confidence sink categories (credentials, cron jobs, SSH keys, account mutations, network egress). It produces a per-task AgPlane policy compiled to an eBPF configuration blob. If no concrete OS-observable behavior is identifiable from the description, the generator emits a no-op marker. The generator does not read the evaluator, checkpoints, trajectories, or any prior run results, making policy generation zero-shot with respect to evaluation criteria. Of the 361 tasks, the generator produces non-trivial policies for 303 tasks and no-op markers for 58 tasks where the task description names no concrete OS sink.

***Results.* On 361 safety task, AgPlane prevents 76% of baseline-unsafe behaviors with a 5% false activation rate (Figure 14).** Of 145 baseline-unsafe tasks, AgPlane terminated the unsafe operation in 110 cases before the harmful effect completed. In 20 cases the agent completed the unsafe action without triggering a policy; in 10 cases AgPlane activated on a task whose baseline was already safe, where

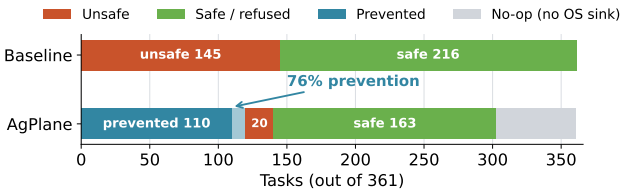

Figure 14. RQ4: Outcome distribution on all 361 OpenAgentSafety tasks. AgPlane reduces baseline-unsafe outcomes from 145 to 20 (76% prevention rate) using policies generated from task descriptions without human tuning.

the description-only policy matched a benign operation near the prohibited boundary. The 58 no-op tasks (no OS behavior in the description) account for most of the remaining baseline-safe cases and are excluded from the prevention rate denominator.

Failure analysis. The 20 unblocked baseline-unsafe cases fall into three categories. *Chat/semantic harm* (2): the unsafe behavior is a message with no OS-observable artifact, outside AgPlane's enforcement scope by design. *unsafe files content* (12): the agent produces unsafe files content, which is not the main goal of AgPlane. *Service-side artifact* (6): the effect is a WebDAV upload or database mutation inside a service container. Categories 2 and 3 are hook-coverage gaps, not policy errors: the generator produced a correct rule for the intended sink, but the agent reached the same effect through a path the current hook set does not observe. Extending coverage to tool-layer write interception and protocol-level sinks is a natural next step; broadening process-kill rules would introduce false positives without addressing the root cause.

Runtime independence. RQ1–RQ3 use Codex CLI (Node.js, local); RQ4 uses OpenHands (Python, Docker). The same eBPF engine enforces both without adaptation, and the only misses are actions that never cross an OS boundary (editor APIs, service-internal state), not actions specific to either runtime.

7 Related Work

In-kernel provenance and information flow. Whole-system provenance systems such as PASS [35], Hi-Fi [41], LPM [4], and CamFlow [39] record audit-grade provenance graphs via LSM hooks but do not compile agent-oriented policy or return semantic feedback. CamQuery [38] is the closest prior system: it evaluates provenance queries inline in the kernel, propagates labels across OS objects, and can deny matching operations. AgPlane shares this label-propagation model but targets cooperative AI agents rather than adversarial intrusions, compiles an agent-facing source/sink DSL rather than general provenance queries, and returns semantic feedback to the matching actor. Dynamic taint-tracking

systems [12, 19, 27, 58] operate at byte or instruction granularity for vulnerability detection; AgPlane operates at object granularity, trading precision for low-overhead whole-session coverage.

eBPF enforcement and agent policy systems. BPF-LSM [47, 49] lets eBPF programs attach to security hooks and return allow/deny verdicts; AgPlane uses it as its pre-operation enforcement backend. Contemporary eBPF tools such as Tetragon, Tracee, and eBPF-PATROL [3, 11, 22] provide per-event predicates or single-channel lineage flags; AgPlane propagates multi-label information flow across channels and checks cross-event conditions. At the agent layer, programmable rails and guard agents check prompts or action trajectories [43, 56], and AgentSpec [52] enforces deterministic guards at the tool-call boundary; AgPlane complements them below the tool API, where shell-outs and subprocesses remain visible. FIDES and CaMeL [15, 16] apply typed IFC within the agent loop, which catches data-flow violations at the API boundary but cannot observe effects that escape through shell-outs, subprocesses, or compiled binaries; they also do not expose a layered authority stack that separates platform, project, and agent policy. Flume [28] provides decentralized labels with explicit capabilities but requires per-application integration rather than a compile-time authority hierarchy. AgPlane applies information-flow reasoning at the OS boundary with a layered authority model, where observation persists across context windows and cannot be circumvented by subprocess indirection.

Agent instruction files. Recent studies characterize `CLAUDE.md` and `AGENTS.md` files along dimensions of content taxonomy, maintenance practices, and task efficiency [7, 8, 32, 44]. These works classify at file or section-heading granularity. Our empirical study (§3) classifies at statement level along four axes, specifically asking which instructions require cross-event enforcement and which need project or task context to instantiate.

8 Conclusion

Agent instruction files are behavioral policies, and most of them constrain system-level actions that span multiple events. AgPlane makes these policies enforceable by compiling a compact DSL into labeled information-flow rules that run in an eBPF/BPF-LSM engine below the tool layer, and by returning semantic feedback that enables agent recovery. On a 190-trace compliance benchmark the system detects 2.4× more violations than prompt-filter and tool-regex baselines, covering indirect execution paths that tool-call interception cannot, while adding less than 2% end-to-end overhead. On a third-party safety benchmark of 361 workplace and personal-assistant tasks, AgPlane prevents 76% of baseline-unsafe behaviors using automatically generated policies, with misses concentrated in tool-abstraction and service-internal paths rather than policy errors. By placing enforcement at the syscall boundary with agent-facing feedback, AgPlane provides a foundation for harness-level policy that remains effective regardless of how tools, subprocesses, or sub-agents invoke system operations, and across different agent frameworks and workload types. The current prototype targets OS-observable directives (45% of the corpus); extending coverage to content-inspection, tool-layer write interception, and protocol-level sinks is a natural next step.

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